LESSON 6.3

Explore Numerical Expressions with Exponents

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write and evaluate numbers in exponent form using a base and a power.

WHY THIS MATTERS You're a sound engineer at a music studio. Each time you flip a switch, the volume doubles. The starting volume is 1 unit. After flipping the switch 3 times, the volume is $2^3 = 2 \times 2 \times 2 = 8$ units — that's 8 times louder! How loud does it get after more flips?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Exponent *Spanish: Exponente*	A small number that tells how many times to multiply a number by itself.	In 9^4, the exponent is 4.
Base *Spanish: Base*	The number that gets multiplied by itself.	In 9^4, the base is 9.
Power *Spanish: Potencia*	A number written with a base and an exponent, like 2^3.	9^4
Evaluate *Spanish: Evaluar*	To find the value by putting numbers in for the letters.	Evaluate 3^4: write 3 × 3 × 3 × 3, then multiply step by step: 9 × 9 = 81
Repeated multiplication *Spanish: Multiplicación repetida*	Multiplying the same factor again and again.	9 × 9 × 9 × 9
Expanded form *Spanish: Forma desarrollada*	A power written out as a product of its factors.	2^3 = 2 × 2 × 2

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Write 9 × 9 × 9 × 9 as a power.

1. The repeated factor is 9, so the base is 9.
2. Count the 9s: there are four, so the exponent is 4.
3. Check: the exponent counts factors, not × signs (there are only 3 × signs).

Answer: 9^4; base 9, exponent 4

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's find 5^2. How many times do we multiply 5 by itself?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In this power, the base is _____ and the exponent is _____ .
- The exponent tells me to use the base as a factor _____ times.
- I read this power as _____ to the _____ power.
- A power shows repeated _____ , not repeated _____ .

Word bank power · base · exponent · factor · repeated multiplication · expanded form · squared · cubed · evaluate · product · read · times

WATCH OUT!

A common mistake

A common mistake in Powers and Exponents is multiplying the base by the exponent instead of using the base as a repeated factor

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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Answer: 9^4; base 9, exponent 4

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression shows 5^3 as repeated multiplication?

(A) 3 × 3 × 3 × 3 × 3
(B) 5 × 5 × 5
(C) 5 × 3
(D) 5 + 5 + 5

STEPS TO TRY
1. In 5^3 the base is 5 and the exponent is 3. Ask: which number repeats, and how many times?
2. The exponent counts the factors. Look for the base written as a factor that many times.
3. Say it out loud: 'use 5 as a factor 3 times.' Count the 5's in each choice — you need exactly three of them.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the value of 4^3?

(A) 7

(B) 12

(C) 43

(D) 64

STEPS TO TRY

1. Look at the small raised 3. That exponent counts how many 4's get multiplied — it is not a number to multiply 4 by.
2. Rewrite the power as repeated multiplication first, then multiply one step at a time.
3. Write 4^3 as $4 \times 4 \times 4$. Multiply the first pair ($4 \times 4 = 16$), then multiply that result by the last 4.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the value of 6^2?

(A) 8

(B) 12

(C) 36

(D) 62

STEPS TO TRY

1. The little 2 is the exponent. It counts how many 6's you multiply — it's not a number to add or multiply by.
2. Turn the power into repeated multiplication before you calculate anything.
3. Rewrite 6^2 as 6×6, then multiply those two factors.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Evaluate: $3 \times (4 + 2)^2$

(A) 36 (B) 54

(C) 108 (D) 144

STEPS TO TRY

1. There are three jobs here: parentheses, an exponent, and multiplication. Order of operations decides which comes first.
2. Do the parentheses first, then square that result, then multiply by 3 last.
3. Start inside: $4 + 2 = 6$. Now the problem becomes 3×6^2. Handle the exponent next.

Rewrite the whole expression each line	What I simplified

1. **Grouping** *Inside ( ) first.*
2. **Exponents** *Then powers.*
3. **× ÷** *Left to right.*
4. **+ −** *Left to right.*

EXPLAIN YOUR THINKING

How does the value change when you increase the exponent by 1? Why?

SENTENCE STARTER When the exponent increases by 1, the value is multiplied by ___ because the base is used as a factor ___ more time.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help